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Computational performance of classical and hybrid root-finding methods for real-world nonlinear models

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Abstract

We offer and discuss a two-point root-finding technique which is a hybrid approach that combines Newton's method with the Secant method. The hybrid version can be represented as

$$x_{n+1} = x_n - \frac{2f(x_n)}{f'(x_n) + \frac{f(x_n) - f(x_{n-1})}{(x_n - x_{n-1})}}.$$

This formulation uses derivative's information along with a finite-difference approximation to the derivatives. We give a local convergence study using standard hypotheses and compare the practical performance of the hybrid method to that of Newton, Secant and the Weerakoon–Fernando method (WFM) on a set of real-world nonlinear equations in engineering and physical applications. Numerical experiments give counts of iterations, counts of evaluations of functions and derivatives, and CPU times along with sensitivity to initial guesses. Results show that the proposed hybrid scheme achieves a desirable balance between convergence rate and robustness in the chosen models, and therefore, is especially applicable in the computational context of the contemporary world, where predictable convergence and stability are vital.

Keywords Root-finding, Newton method, Secant method, Hybrid method, Convergence analysis, Nonlinear equations, Colebrook equation

Mathematics Subject Classification Primary 65H05, Secondary 65D99, 65B99

1 Introduction

The nonlinear equations are natural in a very broad sense: they are needed in fluid mechanics, modeling of electrical systems, chemical kinetics, and behavior of materials. Since not all such equations have closed-form solutions, iterative numerical techniques are a key ingredient to the computation of their roots. The Newton method, the Secant method, Fixed-Point iteration, and higher-order Newton-type variants such as



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the Weerakoon–Fernando method are widely studied iterative techniques due to their simplicity and well-understood convergence behavior [1–3].

The approach by Newton can be especially appealing due to its quadratic local convergence. Its performance however becomes poor in a case where the derivatives are not available or are costly to compute, are poorly conditioned and it may diverge with highly nonlinear or oscillatory functions as well as when derivatives are unavailable or are expensive to compute [4, 5]. The method of Secant, overcomes the necessity of computing derivatives but introduces a finite-difference slope approximation to overcome this and sacrifices the convergence rate [6].

Recent years also have seen a revival in the interest in hybrid and improved iterative schemes which seek to achieve the strength of bracketing strategies, but with a quick local convergence of open methods. Comparative studies show that carefully designed hybrid approaches frequently outperform classical schemes (bisection, secant, Newton–Raphson) in iteration counts and CPU time across standard benchmarks [7, 8].

Furthermore, recent advances in optimal iterative methods and semi-local convergence analysis emphasize the importance of developing algorithms that ensure both stability and efficiency under realistic computational conditions [9]. As a result, hybrid iterative methods have emerged as effective and computationally attractive approaches for solving nonlinear equations, offering a balanced trade-off between fast convergence and improved numerical stability [10, 11].

In order to overcome the drawback of classical Newton-type schemes, researchers have proposed various modified and multi-point iterative methods. These include Newton-type two-point and higher-order variants designed to improve convergence characteristics and stability [12, 13], as well as weighted Newton methods [14] and derivative-free approaches [15]. Such developments reflect ongoing efforts to improve the convergence behavior and computational reliability of iterative root-finding methods [16].

More recently, attention has also turned toward hybrid and predictor–corrector formulations that make use of information from multiple iterates—an approach exemplified by the Weerakoon–Fernando method [3], and inspired by this general direction, the present work proposes a two-point Newton–Secant hybrid structure that combines derivative information with a Secant-type slope correction while preserving a comparable computational framework.

In this work, we introduce a simple yet effective two-point Newton–Secant hybrid iteration that employs a harmonically averaged denominator combining derivative information with a Secant-type slope approximation. The proposed method requires only one derivative evaluation per iteration and utilizes information from two consecutive points. We provide a detailed local convergence analysis that establishes superlinear convergence and demonstrate through numerical experiments that the method offers a favorable compromise between Newton’s fast convergence and the robustness of Secant-like strategies.

From a structural perspective, the proposed iteration differs from classical Newton and Secant methods in the way derivative and two-point information are combined. In particular, it incorporates information from two consecutive approximations through a Secant-type slope term in the denominator. Unlike the pure Secant method, the derivative is not replaced but combined with the two-point slope approximation. This

formulation preserves explicit derivative information while also utilizing data from the previous iterate. Consequently, the scheme represents a direct structural modification of the classical Newton update, designed to balance derivative-based correction with two-point information, without increasing the number of derivative evaluations per iteration.

The main contributions of this paper are:

- (i) Formulation of a two-point Newton–Secant hybrid scheme combining derivative and Secant-type slope information;
- (ii) Derivation of the corresponding local error equation and convergence behavior;
- (iii) Comparative numerical evaluation on representative nonlinear models to assess stability and convergence behavior relative to the Newton, Secant, and Weerakoon–Fernando methods.

2 Related work and mathematical preliminaries

In numerical analysis, classical techniques of solving the nonlinear equations have been widely used. The method by Newton is the most influential because it has quadratic convergence and has been known to be sensitive to initial guess as well as depending on derivative evaluation which has been well reported in the literature in 1995 by Kelley [5]. The Secant method eliminates explicit derivatives, providing superlinear convergence and enhanced practicality in problems with complex or noisy derivatives, and also avoiding explicit derivatives of complicated problems that imply awkward and expensive handling of derivatives such as sparse matrices and nonsmooth terms [6]. Fixed-Point iteration, although fundamental, converges only under contractive mappings and often requires reformulating the nonlinear equation into a suitable iterative form [4].

In order to overcome the drawbacks of these classical schemes, researchers have suggested a number of hybrid and adapted approaches of Newton-type methods. Weerakoon and Fernando [3] introduced a modified Newton-step that improves convergence order while maintaining computational efficiency. Amat et al. [17] investigated families of iterative methods with improved convergence behavior and analyzed their dynamical properties.

Hybrid Newton-Secant methods have also been considered as a possible way of achieving the stability of the Secant method and the convergence rate of the Newton method. Several researchers, including Hueso et al. [18], have proposed modified and hybrid Newton-type schemes to improve convergence and stability. While these methods improve robustness in challenging cases, they often incur additional function or derivative evaluations, increasing overall computational cost compared to lighter-weight hybrid strategies.

Despite these developments, most of existing hybrid methods increase algorithmic complexity or computational cost. Our approach differs by using only one derivative evaluation per iteration and incorporating a Secant-based correction to improve robustness, particularly for functions with derivative instability or poor initial approximations. This positions our method closer to computationally lightweight hybrid approaches while retaining the practical benefits of derivative-based schemes.

The above mentioned iterative schemes are the theoretical and computational basis of solving nonlinear equations. We now recapitulate briefly the classical methods and the representative hybrid approaches, which are relevant to the development and analysis of the proposed method.

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be sufficiently smooth and let α be a simple root of f , i.e. $f(\alpha) = 0$ and $f'(\alpha) \neq 0$. We briefly recall the classical iterations.

2.1 Newton's method

Given an initial guess x_0 close to α , Newton's iteration is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Under regularity hypotheses (sufficient differentiability and $f'(\alpha) \neq 0$), Newton's method converges quadratically.

2.2 Secant method

The Secant method is a two-point derivative-free method:

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}.$$

When iterates are close to the root, the Secant method has order of convergence ≈ 1.618 (the golden ratio) and requires one function evaluation per step.

2.3 Weerakoon-Fernando method

[3] The Weerakoon-Fernando method is an iterative numerical technique to solve non-linear equations $f(x) = 0$ with third-order convergence is defined by

$$x_{n+1} = x_n - \frac{2f(x_n)}{f'(x_n) + f'(y_n)}$$

where

$$y_n = x_n - \frac{f(x_n)}{f'(x_n)}$$

3 Proposed Newton–Secant hybrid method and convergence analysis

We propose the following hybrid iteration:

$$x_{n+1} = x_n - \frac{2f(x_n)}{f'(x_n) + \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}}$$

This uses the exact derivative $f'(x_n)$ and a finite-difference approximation $\frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$ of the derivative based on two recent iterates. The denominator may be viewed as a harmonic-like combination that balances derivative and secant information.

3.1 Motivation and interpretation

The denominator approximates twice the reciprocal of the average of reciprocals (heuristic) and aims to temper the Newton step when the derivative is unreliable or when

the secant slope gives useful corrective information. In particular, when the derivative is expensive or noisy, the secant term provides a stabilizing influence.

3.2 Pseudocode

Require: f, f' , initial guesses x_0, x_1 , tolerance ε , maximum iterations $N_{\max}$, safeguard parameter τ

Ensure: Approximate root x^* or failure flag

```

for  $n = 1$  to  $N_{\max}$  do
  if  $|x_n - x_{n-1}| < \tau$  then
    Return failure (secant slope undefined)
  end if
   $s_n = \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$ 
   $d_n = f'(x_n)$ 
  if  $|d_n + s_n| < \tau$  then
    Return failure (zero denominator safeguard)
  end if
   $x_{n+1} = x_n - \frac{2f(x_n)}{d_n + s_n}$ 
  if  $|x_{n+1} - x_n| < \varepsilon$  then
    return  $x_{n+1}$ 
  end if
end for
Return failure (maximum iterations reached)

```

Algorithm 1 Newton–Secant Hybrid Method

In practical implementation, a small safeguard parameter τ is introduced to prevent division by zero due to floating-point limitations. Since double-precision arithmetic has machine epsilon $\varepsilon_{\text{mach}} \approx 2.22 \times 10^{-16}$, τ is chosen proportional to machine precision, typically $\tau = 100 \varepsilon_{\text{mach}}$, which ensures numerical stability without affecting the convergence characteristics of the method.

3.3 Derivation of the proposed Newton–Secant hybrid method

The proposed method is rooted in the Fundamental Theorem of Calculus. For a function $f \in C^3$ in a neighborhood of a root α , the function value at a point x can be expressed as:

$$f(x) = f(x_n) + \int_{x_n}^x f'(\lambda) d\lambda \quad (1)$$

To determine the next iterate x_{n+1} , we set $f(x_{n+1}) \approx 0$, which yields:

$$0 \approx f(x_n) + \int_{x_n}^{x_{n+1}} f'(\lambda) d\lambda \quad (2)$$

3.4 Trapezoidal approximation

As illustrated in Fig. 1, we approximate the definite integral involved in the equation above by the trapezoidal rule. This averages the slopes at the current iterate x_n and the projected iterate x_{n+1} :

$$\int_{x_n}^{x_{n+1}} f'(\lambda) d\lambda \approx \left(\frac{x_{n+1} - x_n}{2} \right) [f'(x_n) + f'(x_{n+1})] \quad (3)$$

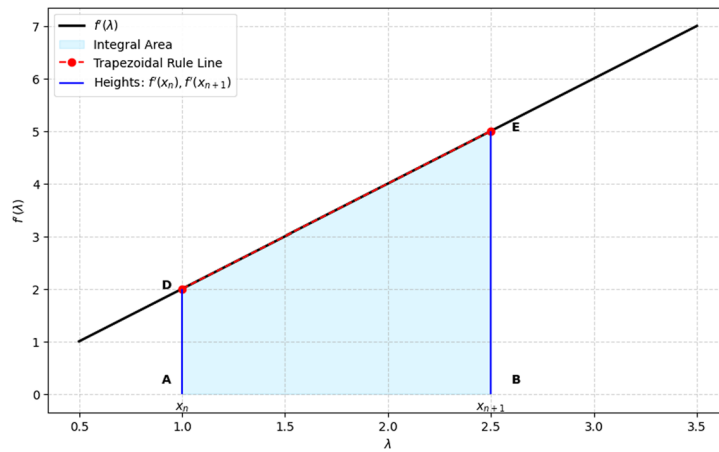


Fig. 1 Geometric interpretation of the trapezoidal approximation. The integral of $f'(\lambda)$ is represented by the area of the trapezoid $ABED$ with heights $f'(x_n)$ and $f'(x_{n+1})$

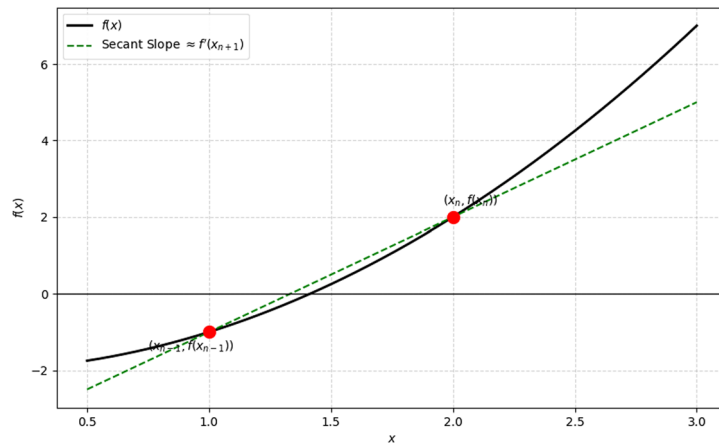


Fig. 2 Secant slope approximation using the two most recent iterates. The derivative at the unknown future iterate x_{n+1} is approximated by the slope of the secant line through $(x_{n-1}, f(x_{n-1}))$ and $(x_n, f(x_n))$

Substituting this approximation back into the root-finding model leads to the implicit iterative scheme:

$$x_{n+1} = x_n - \frac{2f(x_n)}{f'(x_n) + f'(x_{n+1})} \quad (4)$$

3.5 Explicit secant substitution

The iteration above is implicit because the derivative $f'(x_{n+1})$ is evaluated at the unknown next iterate. To transform this into an explicit scheme, we utilize the property of the derivative as the slope of the tangent. As shown in Fig. 2, we approximate the “future” derivative $f'(x_{n+1})$ by the slope of the secant line passing through the two most recent iterates, x_n and x_{n-1} :

$$f'(x_{n+1}) \approx \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}} \quad (5)$$

Substituting this secant approximation into the denominator of the implicit scheme results in the final proposed **Newton-Secant Hybrid Formula**:

$$x_{n+1} = x_n - \frac{2f(x_n)}{f'(x_n) + \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}} \quad (6)$$

This hybrid approach combines derivative's information with a Secant-based slope approximation to obtain an explicit iteration. The resulting method retains the trapezoidal structure while avoiding evaluation of the derivative at the unknown future iterate, requiring only one derivative evaluation per iteration.

3.6 Convergence analysis

We now derive a detailed local convergence analysis of the proposed Newton-Secant hybrid iteration

$$x_{n+1} = x_n - \frac{2f(x_n)}{f'(x_n) + \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}}.$$

Let α be a simple root of f , i.e. $f(\alpha) = 0$ and $f'(\alpha) \neq 0$. Define the error terms

$$e_n = x_n - \alpha.$$

Throughout, assume f is sufficiently smooth (C^3 is enough) in a neighbourhood of α .

3.7 Taylor expansions around the root

Expanding around α , we have:

$$f(x_n) = f'(\alpha)e_n + \frac{f''(\alpha)}{2}e_n^2 + \frac{f'''(\alpha)}{6}e_n^3 + O(e_n^4), \quad (7)$$

$$f'(x_n) = f'(\alpha) + f''(\alpha)e_n + \frac{f'''(\alpha)}{2}e_n^2 + O(e_n^3). \quad (8)$$

Similarly,

$$f(x_{n-1}) = f'(\alpha)e_{n-1} + \frac{f''(\alpha)}{2}e_{n-1}^2 + O(e_{n-1}^3).$$

Define the secant slope

$$s_n = \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}} = \frac{f(x_n) - f(x_{n-1})}{e_n - e_{n-1}}.$$

Using the Taylor expansions, one obtains

$$s_n = f'(\alpha) + \frac{f''(\alpha)}{2}(e_n + e_{n-1}) + O(\max\{e_n^2, e_{n-1}^2\}). \quad (9)$$

3.8 Expansion of the hybrid denominator

The denominator of the hybrid iteration is

$$D_n = f'(x_n) + s_n.$$

Using (8) and (9),

$$D_n = 2f'(\alpha) + f''(\alpha) \left(\frac{3}{2}e_n + \frac{1}{2}e_{n-1} \right) + O(\max\{e_n^2, e_{n-1}^2\}).$$

Thus we may write

$$D_n^{-1} = \frac{1}{2f'(\alpha)} \left[1 - \frac{f''(\alpha)}{2f'(\alpha)} \left(\frac{3}{2}e_n + \frac{1}{2}e_{n-1} \right) \right] + O(\max\{e_n^2, e_{n-1}^2\}).$$

3.9 Expansion of the hybrid step

The iteration can be written as

$$x_{n+1} = x_n - 2f(x_n)D_n^{-1}.$$

Insert the Taylor expansions:

$$x_{n+1} - \alpha = e_n - 2 \left[f'(\alpha)e_n + \frac{f''(\alpha)}{2}e_n^2 + O(e_n^3) \right] \left[\frac{1}{2f'(\alpha)} (1 - C_1(3e_n + e_{n-1})) + O(\max\{e_n^2, e_{n-1}^2\}) \right],$$

where

$$C_1 = \frac{f''(\alpha)}{4f'(\alpha)}.$$

Multiplying out and simplifying,

$$e_{n+1} = C_2e_ne_{n-1} + C_3e_n^2 + O(\max\{e_n^3, e_{n-1}^2\}),$$

A careful coefficient comparison yields the explicit recurrence

$$e_{n+1} = \left(\frac{f''(\alpha)}{2f'(\alpha)} \right) e_n^2 - \left(\frac{f''(\alpha)}{4f'(\alpha)} \right) e_ne_{n-1} + O(e_n^3).$$

Thus,

$$A = \frac{f''(\alpha)}{2f'(\alpha)}, \quad B = -\frac{f''(\alpha)}{4f'(\alpha)}.$$

In particular, if $f''(\alpha) = 0$, then both A and B vanish. In this case the quadratic and mixed terms disappear, and higher-order terms determine the convergence behaviour. Consequently, the method attains at least quadratic local convergence.

3.10 Resulting order of convergence

The recurrence has the general structure

$$e_{n+1} = Ae_n^2 + Be_ne_{n-1} + O(e_n^3).$$

The dominant term in the recurrence is the mixed product e_ne_{n-1} . Assuming asymptotically that

$$e_n \sim Ce_{n-1}^\rho,$$

substitution into the recurrence gives

$$e_{n+1} \sim Ae_{n-1}^{2\rho} + Be_{n-1}^{\rho+1}.$$

For $\rho > 1$, we have $\rho + 1 < 2\rho$, and therefore the mixed term $e_n e_{n-1}$ is asymptotically dominant. Equating the exponents yields the characteristic equation

$$\rho^2 = \rho + 1,$$

whose positive root is

$$\rho = \frac{1 + \sqrt{5}}{2} \approx 1.618.$$

Although the asymptotic convergence order is $\rho \approx 1.618$, finite-precision computations may exhibit small variations in the measured effective order due to rounding effects and the interaction between the quadratic and mixed error terms in the recurrence. However, as demonstrated in Sect. 4, the observed experimental orders remain close to the theoretical golden-ratio value, thereby confirming the validity of the convergence analysis.

Theorem 3.1 (Local Convergence of the Hybrid Method) Let $f \in C^3$ near a simple root α with $f'(\alpha) \neq 0$. Then there exists a neighbourhood $\mathcal{N}(\alpha)$ such that for sufficiently close initial guesses $x_0, x_1 \in \mathcal{N}(\alpha)$, the Newton–Secant hybrid iteration is well-defined and converges to α . The error satisfies

$$e_{n+1} = Ae_n^2 + Be_n e_{n-1} + O(e_n^3),$$

with

$$A = \frac{f''(\alpha)}{2f'(\alpha)}, \quad B = -\frac{f''(\alpha)}{4f'(\alpha)}.$$

If $f''(\alpha) \neq 0$, the method is superlinearly convergent with order

$$\rho = \frac{1 + \sqrt{5}}{2}.$$

If $f''(\alpha) = 0$, the mixed term vanishes and the method attains at least quadratic local convergence.

Proof From the previous expansions, the error satisfies

$$e_{n+1} = Ae_n^2 + Be_n e_{n-1} + O(e_n^3).$$

For sufficiently small e_0, e_1 , the higher-order terms $O(e_n^3)$ remain dominated by the quadratic and mixed terms, and the recurrence reduces asymptotically to a two-step superlinear form. Since the recurrence has the standard two-step form with non-vanishing mixed term $B \neq 0$ (unless $f''(\alpha) = 0$), classical convergence results for multi-step non-linear iterations imply that:

- (1) $e_n \rightarrow 0$ (local convergence);
- (2) $\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|} = 0$ (superlinear convergence);
- (3) Substituting the asymptotic relation $e_n \sim C e_{n-1}^\rho$ into the recurrence gives

$$e_{n+1} \sim A e_{n-1}^{2\rho} + B e_{n-1}^{\rho+1}.$$

For $\rho > 1$, we have $\rho + 1 < 2\rho$, and therefore the mixed term $e_n e_{n-1}$ is asymptotically dominant. Equating exponents on both sides yields

$$\rho^2 = \rho + 1,$$

whose positive root is

$$\rho = \frac{1 + \sqrt{5}}{2}.$$

This completes the proof. $\square$

The convergence region is local and determined by the requirement that $|e_0|$ and $|e_1|$ are sufficiently small so that

- (i) Higher-order terms remain negligible and
- (ii) The denominator $f'(x_n) + s_n$ stays bounded away from zero.

This behaviour is consistent with standard Newton-type methods.

4 Real-world nonlinear models and experimental setup

To evaluate practical performance we test the methods on a suite of nonlinear equations from engineering and applied sciences. The selected models are:

1. $f_1(x) = x^3 - 2x - 5$ (algebraic model)
2. $f_2(x) = \cos x - x$ (transcendental equation arising in fixed points)
3. $f_3(x) = e^{-x} - x$ (chemical/decay balance model)
4. $f_4(x) = x e^x - 5$ (Lambert-W type model)
5. f_5 Colebrook equation for friction factor (implicit in f) – handled via transformation

For each problem we record: number of iterations to tolerance (10^{-10}), total number of function evaluations, derivative evaluations, average CPU time (wall-clock), and success/failure from multiple initial guesses. Convergence rate is estimated numerically using observed errors.

4.1 Experimental setup

To evaluate the performance of the proposed Hybrid Newton–Secant method, we compare it with the following algorithms:

1. **Newton's Method**
2. **Secant Method**
3. **Weerakoon–Fernando Method**
4. **Hybrid Method (proposed)**

All experiments were implemented in Python 3.11 using double-precision floating arithmetic.

Iterations were terminated when

$$|x_{n+1} - x_n| < 10^{-10} \quad \text{or} \quad |f(x_{n+1})| < 10^{-12},$$

or when the maximum of 100 iterations was reached.

The step tolerance 10^{-10} was chosen to provide numerical stability while avoiding unnecessary iterations beyond practically meaningful accuracy in double-precision arithmetic (machine precision $\approx 10^{-16}$). The stricter residual tolerance 10^{-12} is used to assure proper approximation of the true root and allows reliable estimation of the experimental order of convergence.

To prevent uncontrolled divergence, the iteration was terminated if $|x_n| > 10^{10}$. The initial guesses applied to each nonlinear model were fixed and identical across all methods to ensure a fair comparison. Specifically:

- (i) Algebraic equation: $x_0 = 2.0, x_1 = 3.0$
- (ii) Fixed-point equation: $x_0 = 0.5, x_1 = 1.0$
- (iii) Exponential balance equation: $x_0 = 0.5, x_1 = 1.0$
- (iv) Lambert-W type equation: $x_0 = 1.0, x_1 = 2.0$
- (v) Colebrook equation: $x_0 = 0.02, x_1 = 0.03$

For single-initial-point methods, only x_0 was used, while two initial values were used for the Secant and Hybrid methods.

The following performance metrics were collected:

- Number of iterations
- Number of function evaluations (N_f)
- Number of derivative evaluations ($N_{f'}$)
- Total computational cost
- Observed convergence order
- Sensitivity to initial guesses

The computational cost was defined as

$$\text{Cost} = N_f + cN_{f'}.$$

To justify the choice of the weight parameter c , we experimentally measured the CPU time. To reduce the effect of system noise and timing fluctuations, CPU time was averaged over 1000 independent runs for each method and test problem. Derivative evaluations were observed to require approximately three times the CPU time of a function evaluation on average. Therefore, $c = 3$ was adopted as a representative value. For completeness, additional results for $c = 1$ and $c = 2$ are also reported to demonstrate that the comparative ranking of methods remains consistent.

It is worth noting that the parameter c can be adapted depending on application context. In problems where derivatives are inexpensive (e.g., symbolic expressions), smaller values of c may be appropriate, whereas in complex engineering simulations, larger values may better reflect computational effort.

4.2 Test problems

We use five representative real-world nonlinear models:

1. Algebraic Model:

$$f_1(x) = x^3 - 2x - 5.$$

Appears in beam deflection and chemical mixing models.

2. Fixed-Point Model:

$$f_2(x) = \cos x - x.$$

Related to equilibrium conditions and transcendental dynamics.

3. Exponential Balance Equation:

$$f_3(x) = e^{-x} - x.$$

Arises in decay, reaction modelling, and feedback systems.

4. Lambert-W Type Model:

$$f_4(x) = xe^x - 5.$$

Occurs in semiconductor equations and thermal runaway models.

5. Colebrook Equation (Engineering Model):

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\epsilon/D}{3.7} + \frac{2.51}{Re\sqrt{f}} \right),$$

which is implicit in f . We rewrite it as

$$F(f) = \frac{1}{\sqrt{f}} + 2 \log_{10} \left(\frac{\epsilon/D}{3.7} + \frac{2.51}{Re\sqrt{f}} \right) = 0.$$

4.3 Performance comparison and computational cost

Table 1 summarizes the convergence behavior, cost of computation, and accuracy of all methods across the five nonlinear models.

The results confirm that all methods achieve the desired accuracy, while differences in computational cost and CPU time reflect variations in evaluation requirements.

The numerical experiments conducted across a variety of real-world nonlinear models reveal consistent performance trends among the tested methods. Newton's method exhibits fast convergence when the derivative is well conditioned and the initial guess is sufficiently close to the root. The Secant method, although derivative-free and robust, generally requires more iterations across most test problems.

The Weerakoon–Fernando method (WFM), owing to its higher-order convergence, consistently achieves the solution with fewer iterations compared to Newton and Secant

Table 1 Performance comparison of iterative methods on five nonlinear test problems with tolerance 10^{-12}

Function	Method	Iter	N_f	$N_{f'}$	CPU Time (s)	$ f(x^*) $	Root	Cost ($c = 1$)	Cost ($c = 2$)	Cost ($c = 3$)
Algebraic	Newton	5	5	5	5.19×10^{-6}	8.88×10^{-16}	2.094551	10	15	20
	Secant	7	8	0	6.42×10^{-6}	3.55×10^{-15}	2.094551	8	8	8
	Hybrid	7	8	7	9.97×10^{-6}	4.01×10^{-13}	2.094551	15	22	29
	WFM	4	4	8	6.12×10^{-6}	8.88×10^{-16}	2.094551	12	20	28
Fixed-Point	Newton	5	5	5	2.76×10^{-5}	0.00×10^0	0.739085	10	15	20
	Secant	6	7	0	2.31×10^{-5}	3.33×10^{-16}	0.739085	7	7	7
	Hybrid	5	6	5	3.37×10^{-5}	2.55×10^{-15}	0.739085	11	16	21
	WFM	4	4	8	3.31×10^{-5}	1.11×10^{-16}	0.739085	12	20	28
ExpBalance	Newton	4	4	4	2.24×10^{-5}	4.44×10^{-15}	0.567143	8	12	16
	Secant	6	7	0	2.44×10^{-5}	0.00×10^0	0.567143	7	7	7
	Hybrid	6	7	6	4.04×10^{-5}	0.00×10^0	0.567143	13	19	25
	WFM	3	3	6	2.60×10^{-5}	7.77×10^{-16}	0.567143	9	15	21
Lambert-WType	Newton	6	6	6	4.36×10^{-5}	0.00×10^0	1.326725	12	18	24
	Secant	8	9	0	3.15×10^{-5}	1.78×10^{-15}	1.326725	9	9	9
	Hybrid	7	8	7	6.06×10^{-5}	1.96×10^{-13}	1.326725	15	22	29
	WFM	4	4	8	4.93×10^{-5}	5.42×10^{-14}	1.326725	12	20	28
Colebrook	Newton	4	4	4	6.20×10^{-5}	2.46×10^{-11}	0.019005	8	12	16
	Secant	7	8	0	6.58×10^{-5}	0.00×10^0	0.019005	8	8	8
	Hybrid	7	8	7	7.32×10^{-5}	1.78×10^{-15}	0.019005	15	22	29
	WFM	3	3	6	3.52×10^{-5}	5.82×10^{-12}	0.019005	9	15	21

methods, as observed across all test functions. However, each iteration of WFM requires one function evaluation and two derivative evaluations due to its predictor–corrector structure, which influences the overall computational cost depending on the derivative weight factor.

The proposed Hybrid method maintains a favorable balance between convergence speed and robustness. The Hybrid method requires one new function evaluation and one derivative evaluation per iteration after initialization, consistent with its secant-type updating structure. This evaluation count corresponds directly to the algorithmic structure used in the numerical experiments. By integrating derivative-based correction with secant-type information, the method enhances numerical stability while maintaining computational efficiency.

While its iteration counts are slightly higher than Newton’s method in some cases, it demonstrates improved stability and reliability, particularly in problems where derivative sensitivity affects Newton’s performance.

These observations confirm that while higher-order methods such as WFM achieve faster asymptotic convergence, the proposed Hybrid method provides a practical balance between convergence speed, computational cost, and robustness.

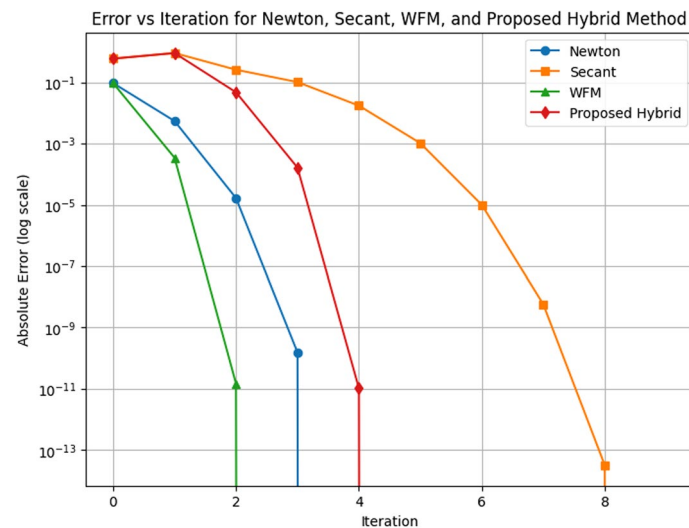
Overall, the Hybrid method consistently outperforms the Secant method in convergence efficiency and provides a competitive and stable alternative to classical derivative-based methods.

4.4 Observed order of convergence

The experimental order of convergence (EOC) was computed using:

Table 2 Experimental observed order of convergence (EOC)

Function	Newton	Secant	Hybrid	WFM
Algebraic	2.014	1.822	1.536	3.022
FixedPoint	2.047	1.490	1.615	3.133
Exp Balance	2.005	1.547	1.486	...
LambertWType	2.060	1.559	1.557	3.113
Colebrook	1.999	1.988	1.515	3.000

**Fig. 3** Error vs iteration for Newton, Secant, WFM, and the proposed hybrid method on the equation $f(x) = x^3 - 2x - 5$

$$\rho_{\text{obs}} = \frac{\ln |e_{n+1}/e_n|}{\ln |e_n/e_{n-1}|}.$$

This gives a numerical verification of the theoretical order from Sect. 3.4.

The experimentally observed orders of convergence for the methods are reported in Table 2.

The EOC value for the WFM in the Exp Balance problem is not reported because the method converged to machine precision rapidly, making the error ratios undefined.

The experimentally computed effective orders are reported in Table 2. The hybrid method consistently exhibits superlinear convergence close to the theoretical golden ratio value.

These results confirm the theoretical convergence properties established in Sect. 3.4. Newton's method demonstrates the expected quadratic convergence, while the Secant method exhibits superlinear convergence consistent with its theoretical order. The Weerakoon–Fernando method converges cubically, as reflected by its experimentally observed order close to three. The proposed Hybrid method consistently maintains stable superlinear convergence across all test problems, validating its theoretical design and demonstrating its reliability as a robust alternative for solving nonlinear equations.

4.5 Convergence behavior

To complement the convergence plot in Fig. 3, we illustrate the convergence behavior of Newton's method, the Secant method, Weerakoon–Fernando method (WFM), and the proposed Hybrid method on the test equation

$$f(x) = x^3 - 2x - 5.$$

Figure 3 illustrates the error reduction behavior of the considered methods. The WFM exhibits the fastest convergence due to its cubic order, followed by Newton and the proposed hybrid method. The Secant method converges more slowly, consistent with its lower convergence order.

In Table 1 the detailed performance metrics, including iteration counts, evaluation counts, CPU time, and computational cost, are reported. The present subsection focuses on illustrating convergence trends through error reduction plots.

Although the proposed method requires both function and derivative evaluations, its hybrid structure improves robustness and reduces sensitivity to instability compared with pure Newton iterations. These observations indicate that the proposed method provides a balanced trade-off between convergence speed and stability.

4.6 Example: Chemical kinetics equation

In this example, we consider a nonlinear equation arising in chemical reaction engineering, particularly in modeling the steady-state relationship between reaction rate, concentration, and kinetic constants. The governing equation is

$$f(x) = 4(1 - e^{-x}) - x = 0,$$

where the parameter value $k = 4$ represents an effective reaction rate constant. This type of equation is typical in the design of continuous stirred tank reactors (CSTRs), catalytic reaction systems, and biochemical kinetics models, where the interplay between exponential decay and linear consumption terms leads to implicit nonlinear equations. Such models rarely admit closed-form solutions, making numerical methods essential.

The purpose of this test is to evaluate the performance and robustness of the proposed hybrid method on a real-world nonlinear equation with a smooth but highly nonlinear structure. Newton's method is expected to perform well due to the smooth derivative, whereas the Secant method may require more iterations. The hybrid method aims to combine the strengths of both approaches by maintaining fast local convergence while reducing sensitivity to initial guesses.

In addition, the Weerakoon–Fernando method (WFM), a third-order method based on a predictor–corrector strategy, is included for comparison to assess the performance of higher-order methods on this nonlinear chemical kinetics model.

The convergence plot and iteration table demonstrate that the hybrid method performs comparably to Newton's method while outperforming the Secant method in terms of the number of iterations required to reach the prescribed tolerance.

Furthermore, the WFM method achieves the fastest convergence among the considered methods, confirming the effectiveness of higher-order predictor–corrector schemes for smooth nonlinear equations.

In Fig. 4, the convergence trends for the four methods applied to this problem are illustrated, which shows the error reduction across iterations.

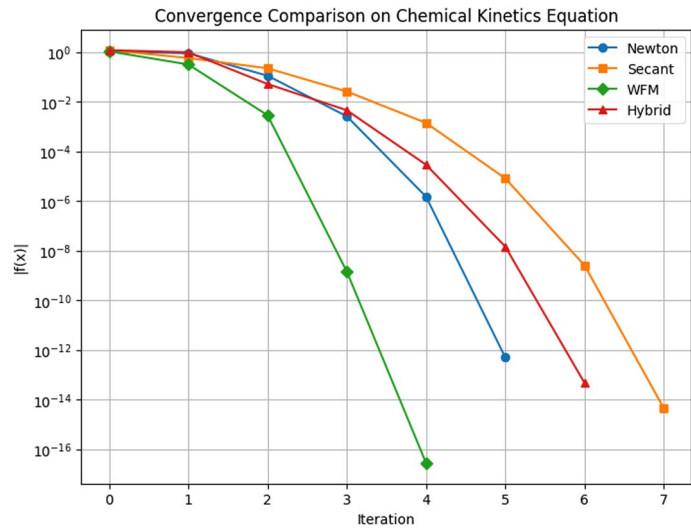


Fig. 4 Convergence behaviour of Newton, Secant, WFM, and Hybrid methods applied to the nonlinear chemical kinetics equation $f(x) = 4(1 - e^{-x}) - x$. The hybrid method shows robustness similar to Newton while outperforming the Secant method. The WFM method demonstrates the fastest convergence due to its higher-order formulation

Table 3 Iteration counts for the chemical kinetics equation $f(x) = 4(1 - e^{-x}) - x$

Method	Iterations
Newton	6
Secant	8
Hybrid	7
WFM	5

As shown in Table 3, Newton’s method required 6 iterations, the Secant method 8 iterations, the Hybrid method 7 iterations, and the WFM method required only 5 iterations, achieving the fastest convergence among all tested methods.

4.7 Robustness analysis under distinct instability mechanisms

This robustness analysis complements the standard efficiency comparison presented in the Experimental Setup subsection. Although the previous experiments used fixed initial guesses to ensure fair efficiency comparison, the present analysis deliberately employs challenging initial conditions to evaluate stability under instability mechanisms.

Three nonlinear test equations are taken in order to determine the stability and reliability of the proposed hybrid method. Each example represents a different instability mechanism that are common in nonlinear root-finding problems:

1. Strong algebraic nonlinearity,
2. Derivative-zero instability,
3. Asymptotic blow-up behavior.

The choice of the initial guesses x_0 were deliberately made to probe different instability regimes of each test equation. Instead of using purely random values, the initial points were chosen in neighborhoods where classical methods are known to exhibit sensitivity, including regions near derivative stationary points, near-flat slopes, and (for transcendental cases) near asymptotic regions. Symmetric perturbations around critical points

were also considered to assess stability under small variations of the initial approximation. This structured selection strategy enables a controlled and reproducible robustness assessment across all methods.

For single-point methods (Newton and WFM), only the initial guess x_0 is required. However, for two-point methods (Secant and Hybrid), the convergence behavior is known to depend on the choice of the initial pair (x_0, x_1) . Therefore, in Case II and Case III, instead of using a single deterministic choice, multiple secondary initial guesses were considered for each x_0 , including:

$$x_1 = x_0 + 0.1, \quad x_1 = x_0 - 0.1, \quad \text{and} \quad x_1 = x_0 + \delta,$$

where $\delta \sim \mathcal{U}(-0.2, 0.2)$ is a uniformly distributed random perturbation. This allows assessment of sensitivity with respect to the secondary initial guess while maintaining reproducibility of the experimental framework.

For Case I, a single deterministic choice of $x_1 = x_0 + 0.1$ was used to maintain consistency with the standard experimental setup, since the observed behavior in this case did not exhibit significant sensitivity to variations in the secondary initial guess.

The above three choices provide a representative combination of deterministic and randomized perturbations, enabling assessment of sensitivity to the secondary initial guess without introducing excessive computational overhead.

All methods are tested under identical stopping criteria:

$$|x_{n+1} - x_n| < 10^{-12},$$

with a maximum of 50 iterations. The tolerance 10^{-12} is chosen to ensure high accuracy relative to double-precision arithmetic ($\approx 10^{-16}$), while avoiding premature termination due to round-off effects.

The following status classifications are used:

- **Success:** convergence within tolerance,
- **DivZero:** denominator collapse,
- **MaxIter:** maximum iteration limit reached,
- **Divergence:** iterates grow unbounded.

4.7.1 Case I: Algebraic nonlinear equation

Consider the algebraic nonlinear equation

$$f(x) = x^3 - 2x + 2.$$

This example evaluates the sensitivity of open methods to basin-of-attraction effects. Although the equation possesses a unique real root at $x \approx -1.769292$, initial guesses near the origin lead to oscillatory behavior and failure of certain methods to enter the correct convergence region (Table 4).

Newton's method fails to converge for initial guesses near the origin. The iterates stagnate close to zero without satisfying the prescribed residual tolerance, ultimately reaching the maximum iteration limit. Since $f(0) \neq 0$, this behavior reflects basin-of-attraction instability rather than derivative singularity. The Secant and WFM methods exhibit partial stability; however, the proposed hybrid method consistently converges to

Table 4 Performance comparison for the nonlinear equation $x^3 - 2x + 2$

S.No	x_0	Newton			Secant			WFM			Hybrid		
		Iter	Approx	Status	Iter	Approx	Status	Iter	Status	Approx	Iter	Approx	Status
1	-0.100	50	0.000000	MI	50	-1.653164	MI	50	MI	0.818797	21	-1.769292	S
2	-0.050	50	0.000000	MI	35	-1.769292	S	21	S	-1.769292	17	-1.769292	S
3	-0.005	50	0.000000	MI	39	-1.769292	S	50	S	-0.806076	14	-1.769292	S
4	0.000	50	0.000000	MI	50	-4.587729	MI	28	MI	-1.769292	14	-1.769292	S
5	0.005	50	0.000000	MI	34	-1.769292	S	50	S	0.818049	13	-1.769292	S
6	0.050	50	0.000000	MI	31	-1.769292	S	22	S	-1.769292	11	-1.769292	S
7	0.100	50	0.000000	MI	40	-1.769292	S	10	S	-1.769292	9	-1.769292	S

S – Successful convergence; MI – Maximum iterations reached

the real root $x \approx -1.769292$ within fewer iterations, demonstrating superior robustness under algebraic nonlinearity.

4.7.2 Case II: Derivative-zero instability

Consider the transcendental equation

$$f(x) = \log(1 + x^2) - 1,$$

with derivative

$$f'(x) = \frac{2x}{1 + x^2}.$$

Since $f'(1) = 0$, derivative-based schemes may experience denominator collapse near this point. This example illustrates instability arising from near-zero slopes (Table 5).

For the nonlinear equation $\log(1 + x^2) - 1 = 0$, the derivative vanishes at $x = 0$, causing structural failure (DZ) in Newton and WFM due to division by zero. For small initial guesses near zero, Newton and WFM further exhibit divergence (DG) characterized by rapidly growing iterates, reflecting instability of derivative-based schemes in regions where $f'(x)$ is small.

In contrast, the Secant and Hybrid methods exhibit comparatively stable behavior in most tested cases and converge to one of the two symmetric roots $x \approx \pm 1.310832$, depending on the choice of the secondary initial guess x_1 . This indicates that convergence is influenced by the initial pair (x_0, x_1) , and may vary when both points lie in regions with unfavorable slope conditions.

These results highlight the sensitivity of derivative-based methods near stationary points, while indicating that the performance of two-point methods depends on the selection of initial guess selection.

4.7.3 Case III: Asymptotic instability

Consider the transcendental equation

$$f(x) = \tan x - x,$$

with derivative

$$f'(x) = \sec^2 x - 1 = \tan^2 x.$$

The function $\tan x$ possesses vertical asymptotes at

$$x = (2k + 1) \frac{\pi}{2}, \quad k \in \mathbb{Z},$$

where both $f(x)$ and $f'(x)$ become unbounded. Consequently, in neighborhoods of these asymptotes the nonlinear equation exhibits severe numerical sensitivity (Table 6).

When the initial guess is near an asymptote, the derivative term $\sec^2 x$ becomes excessively large, leading to unstable updates in Newton-type methods. Consequently, both Newton and the Weerakoon–Fernando method often produce very large iterates and diverge.

In contrast, the Secant method avoids direct derivative evaluation and therefore exhibits comparatively stable behavior across the tested initial pairs. However, the

Table 5 Performance analysis for $f(x) = \log(1 + x^2) - 1$ with multiple initial pairs (x_0, x_1)

x_0	x_1	Newton			Secant			Hybrid			WFM		
		Iter	Final	Status	Iter	Final	Status	Iter	Final	Status	Iter	Final	Status
-0.006	0.094000	10	1.579101 $\times 10^{10}$	D	10	1.310832	S	9	1.310832	S	8	1.883467 $\times 10^{11}$	D
	-0.106000	10	1.579101 $\times 10^{10}$	D	8	-1.310832	S	8	-1.310832	S	8	1.883467 $\times 10^{11}$	D
	0.070510	10	1.579101 $\times 10^{10}$	D	11	-1.310832	S	8	-1.310832	S	8	1.883467 $\times 10^{11}$	D
-0.001	0.099000	9	-8.403777 $\times 10^{10}$	D	9	1.310832	S	8	1.310832	S	7	-7.556437 $\times 10^{10}$	D
	-0.101000	9	-8.403777 $\times 10^{10}$	D	8	-1.310832	S	8	-1.310832	S	7	-7.556437 $\times 10^{10}$	D
	0.136658	9	-8.403777 $\times 10^{10}$	D	8	1.310832	S	7	1.310832	S	7	-7.556437 $\times 10^{10}$	D
0.000	0.100000	0	0.000000	DZ	8	1.310832	S	8	1.310832	S	0	0.000000	DZ
	-0.100000	0	0.000000	DZ	8	-1.310832	S	8	-1.310832	S	0	0.000000	DZ
	-0.000252	0	0.000000	DZ	11	7.029744 $\times 10^{10}$	D	9	-1.085634 $\times 10^{10}$	D	0	0.000000	DZ
0.001	0.101000	9	8.403777 $\times 10^{10}$	D	8	1.310832	S	8	1.310832	S	7	7.556437 $\times 10^{10}$	D
	-0.099000	9	8.403777 $\times 10^{10}$	D	9	-1.310832	S	8	-1.310832	S	7	7.556437 $\times 10^{10}$	D
	0.076199	9	8.403777 $\times 10^{10}$	D	10	-1.310832	S	9	-1.310832	S	7	7.556437 $\times 10^{10}$	D
0.060	0.160000	6	1.310832	S	7	1.310832	S	7	1.310832	S	9	-8.661167 $\times 10^{10}$	D
	-0.040000	6	1.310832	S	15	-8.287156 $\times 10^{10}$	D	14	4.845533 $\times 10^{10}$	D	9	-8.661167 $\times 10^{10}$	D
	-0.004357	6	1.310832	S	16	4.868512 $\times 10^{10}$	D	13	1.372684 $\times 10^{10}$	D	9	-8.661167 $\times 10^{10}$	D

Notation: S = Success (convergence to a valid root), D: Diverged DZ: Division by zero

Table 6 Performance comparison for $f(x) = \tan x - x$ under Multiple Initial Pairs

x_0	x_1	Newton			Secant			Hybrid			WFM		
		Iter	Final	Status	Iter	Final	Status	Iter	Final	Status	Iter	Final	Status
-7.860	-7.7600	24	-1.1556 $\times 10^{10}$	DG	8	-7.725252	S	7	-7.725252	S	34	-1.7752 $\times 10^{10}$	DG
	-7.9600	24	-1.1556 $\times 10^{10}$	DG	42	-0.000114	S	38	1.7783 $\times 10^{10}$	DG	34	-1.7752 $\times 10^{10}$	DG
	-7.8914	24	-1.1556 $\times 10^{10}$	DG	44	0.000143	S	32	-7.9617 $\times 10^{11}$	DG	34	-1.7752 $\times 10^{10}$	DG
-7.880	-7.7800	16	-2.1578 $\times 10^{10}$	DG	9	-7.725252	S	9	-7.725252	S	28	-2.3904 $\times 10^{10}$	DG
	-7.9800	16	-2.1578 $\times 10^{10}$	DG	37	0.000111	S	13	-1.0904 $\times 10^1$	DG	28	-2.3904 $\times 10^{10}$	DG
	-7.9783	16	-2.1578 $\times 10^{10}$	DG	42	0.000126	S	44	-1.3246 $\times 10^{10}$	DG	28	-2.3904 $\times 10^{10}$	DG
-7.900	-7.8000	20	-1.1683 $\times 10^{11}$	DG	10	-7.725252	S	9	-7.725252	S	31	-1.3591 $\times 10^{10}$	DG
	-8.0000	20	-1.1683 $\times 10^{11}$	DG	42	-0.000111	S	32	-1.4283 $\times 10^{10}$	DG	31	-1.3591 $\times 10^{10}$	DG
	-7.8104	20	-1.1683 $\times 10^{11}$	DG	11	-7.725252	S	32	2.0783 $\times 10^{10}$	DG	31	-1.3591 $\times 10^{10}$	DG
-7.857	-7.7570	23	-6.4304 $\times 10^{10}$	DG	8	-7.725252	S	7	-7.725252	S	43	-2.5510 $\times 10^{10}$	DG
	-7.9570	23	-6.4304 $\times 10^{10}$	DG	16	-4.493409	S	41	2.9928 $\times 10^{10}$	DG	43	-2.5510 $\times 10^{10}$	DG
	-7.8327	23	-6.4304 $\times 10^{10}$	DG	12	-7.725252	S	11	-7.725252	S	43	-2.5510 $\times 10^{10}$	DG
-7.800	-7.7000	6	-7.725252	S	7	-7.725252	S	6	-7.725252	S	4	-7.725252	S
	-7.9000	6	-7.725252	S	15	-7.725252	S	32	-2.0538 $\times 10^{10}$	DG	4	-7.725252	S
	-7.8184	6	-7.725252	S	9	-7.725252	S	9	-7.725252	S	4	-7.725252	S

Notation: S = Success (convergence to a valid root), DG = Diverged

convergence is not restricted to a single root. Instead, the Secant method converges to different valid roots, including $x \approx -7.725252$, $x \approx -4.493409$, and $x \approx 0$, depending on the choice of the initial pair (x_0, x_1) .

The proposed hybrid method demonstrates mixed behavior: for certain initial pairs, it successfully converges to the principal root $x \approx -7.725252$, while for others it produces large iterates and diverges.

These observations indicate that, although the hybrid correction can improve stability in some cases, its performance remains sensitive to the choice of initial guesses, particularly in the vicinity of asymptotic regions.

Thus, this case highlights the asymptotic instability of derivative-dependent schemes, while also illustrating the dependence of two-point methods on initial pair selection.

4.7.4 Overall observation

The robustness investigation confirms that classical derivative-based methods exhibit instability under derivative-zero and asymptotic conditions. Across all three instability mechanisms, the proposed hybrid approach demonstrates improved stability in several cases, although its convergence behavior is not uniform and depends on the choice of initial guesses.

5 Discussion, conclusion with future work

The above numerical tests provide the realistic dynamics of the proposed hybrid system, i.e. the Newton - Secant method. We also include the Weerakoon–Fernando Method (WFM), a well-established third-order iterative method, as a benchmark for comparison with the proposed hybrid method. The hybrid scheme is not always a superior scheme in terms of the iterations required or in absolute rate of convergence, however the overall performance of the hybrid scheme shows remarkable and steady trend thus making it worthy to be used in solving nonlinear problems obtained as a result of practical applications.

5.1 Comparison with existing methods

Classical methods have been compared to the findings of the current study. In all test problems, Newton converged faster than any other method of converging a problem, provided the derivative was well-behaved, and the initial approximation was close enough to the solution, as is expected in its quadratic convergence property. The Weerakoon–Fernando method (WFM), owing to its third-order convergence, often achieved faster convergence than Newton's method when derivative evaluation remained stable and well-defined. The Secant method does not require derivative evaluation and usually needs more iterations, but it performs better when the derivative is small, ill-conditioned, or difficult to compute.

The suggested hybrid approach exhibited reliable superlinear convergence across all test problems while reducing the divergence and instability occasionally observed in Newton iterations. By combining derivative information with a secant-based correction, the method improves robustness while maintaining efficient convergence. The Weerakoon–Fernando Method (WFM) was included as a higher-order benchmark for comparison. The results indicate that the proposed hybrid method maintains stable convergence and competitive computational performance across the nonlinear test problems.

5.2 Stability and practical robustness

Certainly, practical robustness contributes significantly to the stability of the system. One significant advantage of the hybrid method is its improved stability in certain nonlinear equations where Newton's iteration may overshoot or diverge, particularly in regions with nearly flat or rapidly changing derivatives. In such situations, the secant correction controls the update step and prevents excessively large iterations.

This behavior was reflected in problems involving exponential nonlinearities and chemical kinetics models, where derivative behavior is highly nonlinear and changes more quickly. Compared to the classical Secant method, the hybrid scheme often converges efficiently while maintaining stable behavior; however, its performance can vary depending on the initial pair selection.

Similarly, although the WFM method demonstrates faster convergence in smooth cases due to its higher-order formulation, the hybrid method provides a balance between stability and computational simplicity, but does not uniformly outperform existing methods under all initial conditions.

5.3 Computational trade-off

The numerical results indicate an evident trade-off between speed and reliability. Newton's method is very efficient but sensitive to poor initial guesses or problematic derivatives, whereas the Secant method is more robust but slower. The WFM method generally requires fewer iterations in smooth problems due to its predictor–corrector formulation, as this advantage is accompanied by additional function evaluations per iteration. The hybrid approach lies between these methods: it typically converges faster than the Secant method while avoiding many of the instability issues associated with Newton's iteration. The additional computational cost remains moderate, as the algorithm requires one derivative evaluation and one secant slope per iteration.

5.4 Practical implications and limitations

Regarding the implementation aspect, the hybrid method offers several advantages. Because the method is less sensitive to derivative behavior, it is well-suited to problems where analytical derivatives are difficult or expensive to compute. The moderated update step often leads to better convergence in engineering and scientific applications, particularly when the initial guesses are far from the solution.

However, certain limitations should be acknowledged. Evaluation of a derivative is necessary at every iteration, which may restrict applicability in black-box simulations. The algorithm also employs two-point memory, introducing a small storage overhead. In addition, for highly oscillatory functions where derivatives frequently approach zero, careful initialization remains necessary.

Overall, the proposed hybrid Newton–Secant algorithm is a reliable and computationally efficient alternative to classical root-finding methods, especially in the context of real-world models in which stability is as important as the convergence rate. The comparative results with Newton, Secant, and WFM methods confirm that the hybrid scheme provides a favorable balance between convergence speed, robustness, and computational cost for the problems examined.

5.5 Conclusion with future work

This study described a simple, but practically useful, two-point Newton–Secant hybrid root-finding method and examined its convergence, stability, and computational efficiency. Comparative evaluation with Newton, Secant, and Weerakoon–Fernando methods provides a comprehensive assessment of its practical performance. The novelty of the method lies in the harmonically averaged correction step, as it provides a balanced compromise between the fast local convergence of Newton’s method and the robustness of Secant-type iterations. In contrast to a number of earlier hybrid variants, the proposed approach shows reliable performance on real-world nonlinear models, including engineering flow equations, exponential–transcendental systems, and implicit relations involving the Lambert–W function.

The key contributions of the work may be summarized in the following way:

1. *Hybrid Formulation*: A new correction strategy combining derivative-based and derivative-free slopes is introduced, improving global behavior while retaining local efficiency.
2. *Convergence Analysis*: Theoretical analysis establishes superlinear convergence with order $\rho \approx 1.618$. The experimental order of convergence (EOC) computed numerically ranges from 1.4 to 1.615, which is consistent with the theoretical prediction.
3. *Numerical Validation*: Extensive experiments on algebraic, transcendental, and engineering models show that the hybrid method provides reliable and stable convergence behavior. Although higher-order methods such as WFM may converge faster in ideal smooth cases, the hybrid method offers a favorable balance between robustness, computational cost, and stability, particularly in problems involving instability mechanisms or sensitive initial conditions.

Future Work:

- Extension of the proposed strategy to multidimensional nonlinear systems using Jacobian–Secant hybridizations.
- Development of adaptive weighting mechanisms to further enhance global convergence properties.
- Design of derivative-free variants suitable for black-box or simulation-based models.
- Investigation of hybrid combinations with higher-order predictor–corrector schemes to further improve convergence efficiency while maintaining robustness.
- Integration of the hybrid scheme into large-scale simulation and optimization solvers to assess real-time computational performance.

In conclusion, the proposed Hybrid Newton–Secant method provides a numerically stable and computationally efficient alternative to classical root-finding techniques. Its balanced performance relative to Newton, Secant, and WFM methods, combined with strong robustness characteristics, makes it a useful addition to existing iterative solution methods.

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Author contributions

R.S. and V.S.C. conducted a research and investigation process, specifically performing the experiments, or data/evidence collection.

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Data availability

All numerical data generated and analysed during this study are included within this published article. No external datasets were used, and no additional data are required to reproduce the findings.

Declarations**Ethics approval and consent to participate**

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Consent for publication

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Competing interests

The authors declare no competing interests.

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